LIBRA Liquid Immersion Blanket - Robust (tritium) Accountancy

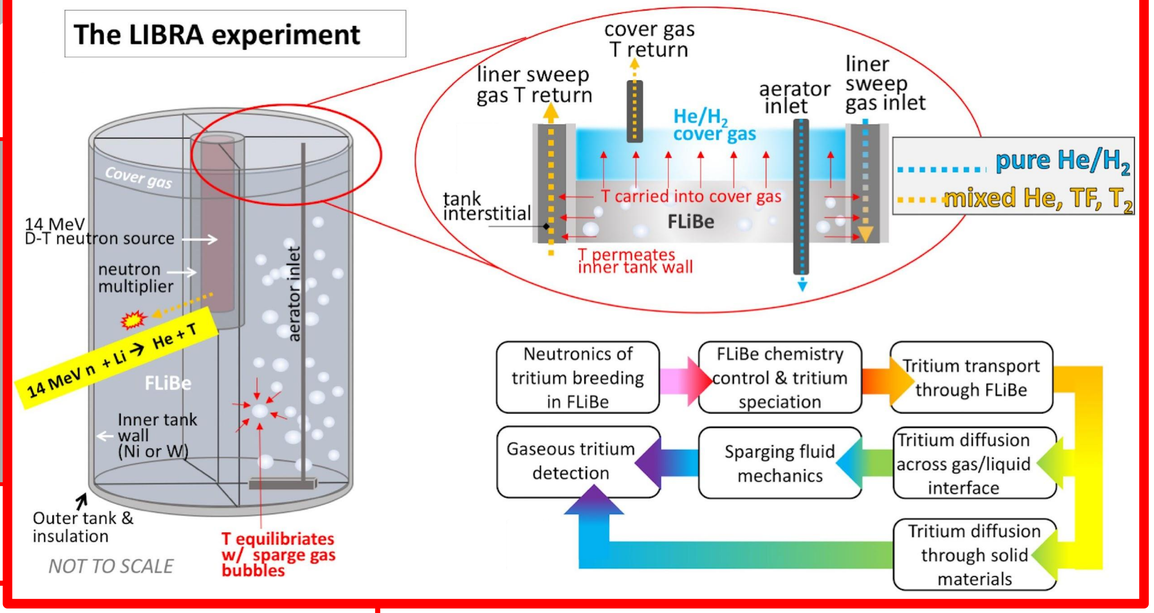
ARPAE AWARDEE **OPEN 2021**

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The LIBRA facility

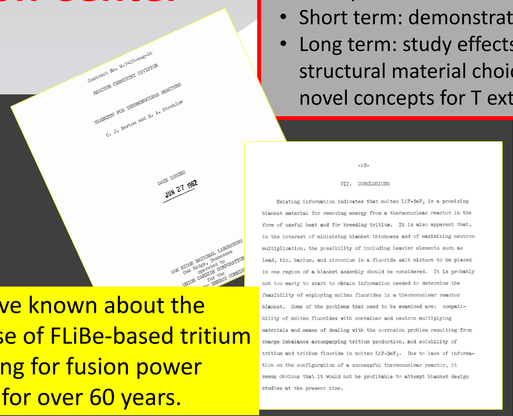
- D-T neutron generator in a tank of molten FLiBe gives us tritium breeding in a LIB-relevant neutron spectrum
- 4 quadrant tanks operate independently, allowing for repeatability and parametric studies
- Constructing a new lab space at MIT PSFC specifically for experiments that use, handle, and purify FLiBe
- Short term: demonstrate global TBR measurements
- Long term: study effects of salt chemistry and structural material choices on TBR / T extraction; test novel concepts for T extraction and control

The LIBRA experiment

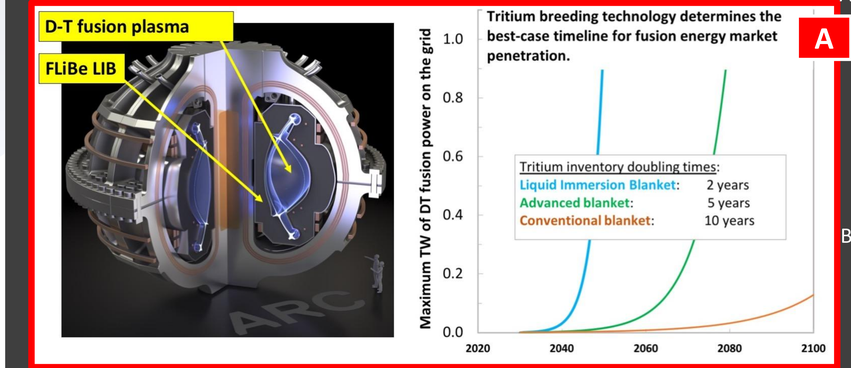


Why FLiBe? Why a liquid immersion blanket (LIB)?

- Low-Z elements & neutron multiplication from Be : good moderation (and magnet shielding), high flux
- Maximize volume of breeding material
- FLiBe also functions as the working fluid / coolant
- Simple design is cheaper to build and minimizes structural material that activates, corrodes, and/or parasitically absorbs neutrons
- Easy to drain to allow for maintenance
- FLiBe is a lifetime component



We have known about the promise of FLiBe-based tritium breeding for fusion power plants for over 60 years.

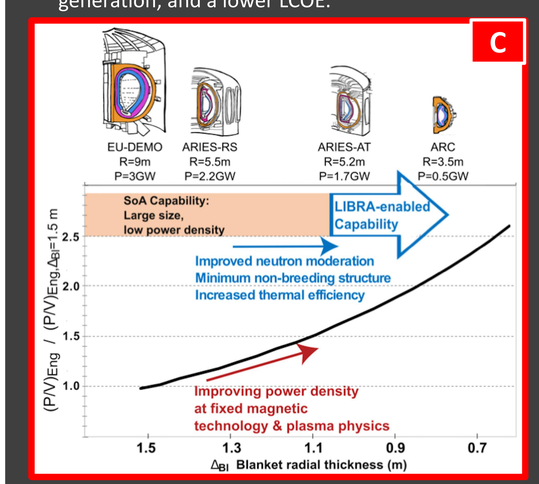


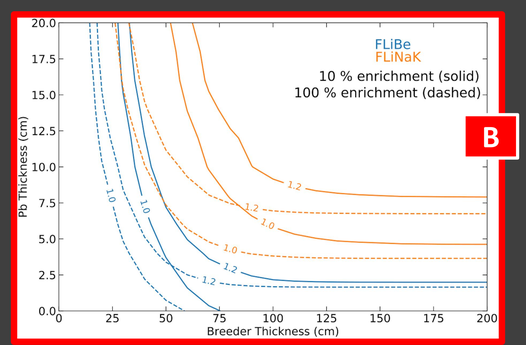
A. The LIB surrounds the vacuum vessel and plasma in the proposed ARC design being developed by CFS. Higher TBR and efficient extraction means more fusion power on the grid faster.

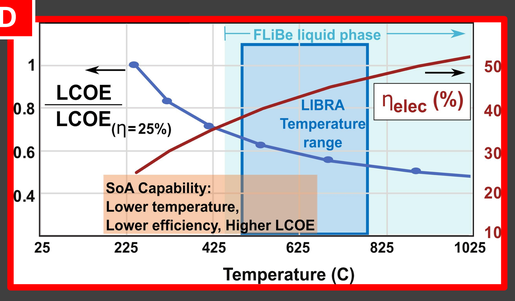
B. FLiNaK is easier to handle (no beryllium) but would require a much larger blanket radial build to achieve the same TBR.

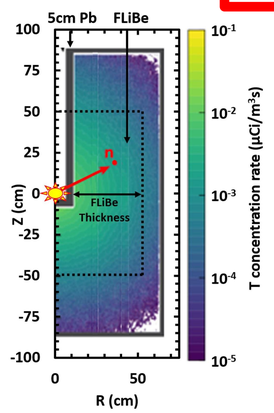
C. The LIB enables a smaller radial build and a more economic fusion power plant. LIBRA will be the first scaled-up proof-of-concept of the proposed LIB technology.

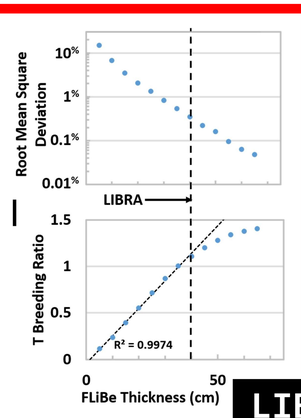
D. The FLiBe LIB allows us to reach higher temperatures, higher efficiency of power generation, and a lower LCOE.











Neutronics inform the design of LIBRA.

T production is concentrated near the neutron source (here, modeled using OpenMC at 10^{10} n/s). LIBRA's dimensions were chosen to maximize tritium breeding and fidelity to the LIB breeding environment while keeping the tank and FLiBe volume to a manageable size. We targeted a RMS deviation in neutron energy spectrum (relative to a full-scale LIB) that was <1% and a TBR close to 1. LIBRA will have a radial FLiBe thickness of ≈40 cm. Larger tank sizes would gain only marginal improvements in TBR.

LIBRA is a bold step forward in experimental studies of FLiBe tritium breeding.

Intrexflibe (Japan) and JUPITER-II (Japan-US) are the most comprehensive FLiBe tritium breeding campaigns to date, but there are major gaps in our knowledge. We need to study T accountancy, kinetics, and FLiBe chemistry in large volumes of FLiBe and in a 14 MeV neutron spectrum, and we need to do it soon in order to deploy the LIB on the timescale proposed by commercial fusion ventures.

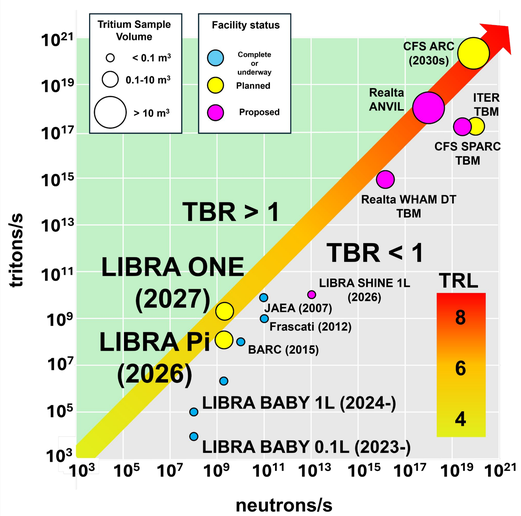
Note: for citation information, see our upcoming paper in Fusion Science & Technology or email seferry@mit.edu.

LIBRA□		Neutrons	14 MeV fusion spectrum	Tritium	Measure TBR	Complementary neutronics	T extraction, detection, & speciation	T transport in FliBe	T diffusion at gas/liquid IF	T diffusion through solids	Redox control	Material compatibility	Sparging fluid effects	> 1 L FliBe
		Radiation		T accountancy					T kinetics		FLiBe chemistry			
INTREXFLIBE [68]	[99]	Y	Y	Y	N	N	Y	N	Y	N	N	Y	Y	N
	[94]	Y	N	Y	N	N	Y	N	N	N	N	N	N	N
	[101]	Y	N	Y	N	N	Y	Y	Y	N	N	N	Y	N
	[102, 103]	Y	N	Y	N	N	Y	N	Y	N	N	N	Y	N
	[105]	Y	N	Y	N	N	Y	N	Y	Y	N	N	N	N
	[118]	N	N	N	N	N	N	N	N	N	N	Y	N	N
JUPITER II [97]	[107, 108]	N	N	N	N	N	N	D, H used as sub. for T			N	N	N	N
	[112]	N	N	Y	N	N	Y	Y	N	N	N	N	N	N
	[89]	Y	N	Y	N	Y	Y	Y	N	N	N	N	N	N
Redox experiments	[120-122]	N	N	N	N	N	N	N	N	N	Y	N	N	N
	[123]	N	N	N	N	N	N	N	N	N	Y	Y	N	N
FLiBe corrosion studies	[125, 126]	N	N	N	N	N	N	N	N	N	N	Y	N	~
	[133]	Y	N	Y*	N	N	N	N	N	N	N	Y	N	N
	[104, 129-132, 134, 136-138]	N	N	N	N	N	N	N	N	N	N	Y	N	N

Non-experimental papers or papers using FLiNaK were not included.

* FLiBe was Li-7 enriched to reduce T production

~ Loop tests with flowing FLiBe.



Level	NASA TRL definition for spaceflight technologies	Applying the TRL framework to the LIB
TRL 9	System "flight proven" through successful mission operations	Many LIB-equipped FPPs are operating successfully
TRL 8	Actual system completed and "flight qualified" through demonstration	T breeding and extraction demonstrated on an FPP with the first full-scale LIB
TRL 7	System prototype demonstration in a space environment	LIB test blanket module is deployed and tested on a DT-tokamak with Q>1
TRL 6	System prototype demonstration in a relevant environment	LIBRA PHASE II: scale up the system
TRL 5	Component/breadboard validation in a relevant environment	LIBRA PHASE I : demonstrate global TBR > 1 in a radiochemically relevant environment
TRL 4	Component/breadboard validation in a laboratory environment	T successfully extracted from FLiBe following neutron irradiation (INTREXFLIBE, JUPITER-II, etc.)
TRL 3	Proof-of-concept demonstrated	Analysis demonstrates validity of LIB concept (LIB/ARC introduced in 2015 (Sorbom et al.))
TRL 2	Technology concept formulated	The LIB concept is created
TRL 1	Basic principles observed and reported	Interactions between neutrons and the lithium in FLiBe create tritium

We have to rapidly advance the LIB's TRL.

Tritium breeding blanket designs are still largely conceptual, and most tritium breeding experiments have been small-scale tests of breeding material in a fission neutron environment. Only a few tritium breeding experiments – none of which used FLiBe - have used fusion neutrons (plotted here). Only the OKTAVIAN experiment achieved a global TBR ≈ 1, but this was is an idealized geometry designed for cross-section measurements, not the validation of a blanket design. Designs like ARC call for a FLiBe LIB and target construction and operation within ten years, but the LIB is currently only at a TRL 2-3.